Name
Date
Course

Transforming the Graphs of Trigonometric Functions

General Forms of Trigonometric Functions

Sinusoidal functions.

$y = A\sin(\omega x - \phi) + B$

$y = A\cos(\omega x - \phi) + B$

$y = A\csc(\omega x - \phi) + B$

$y = A\sec(\omega x - \phi) + B$

$y = A\tan(\omega x - \phi) + B$

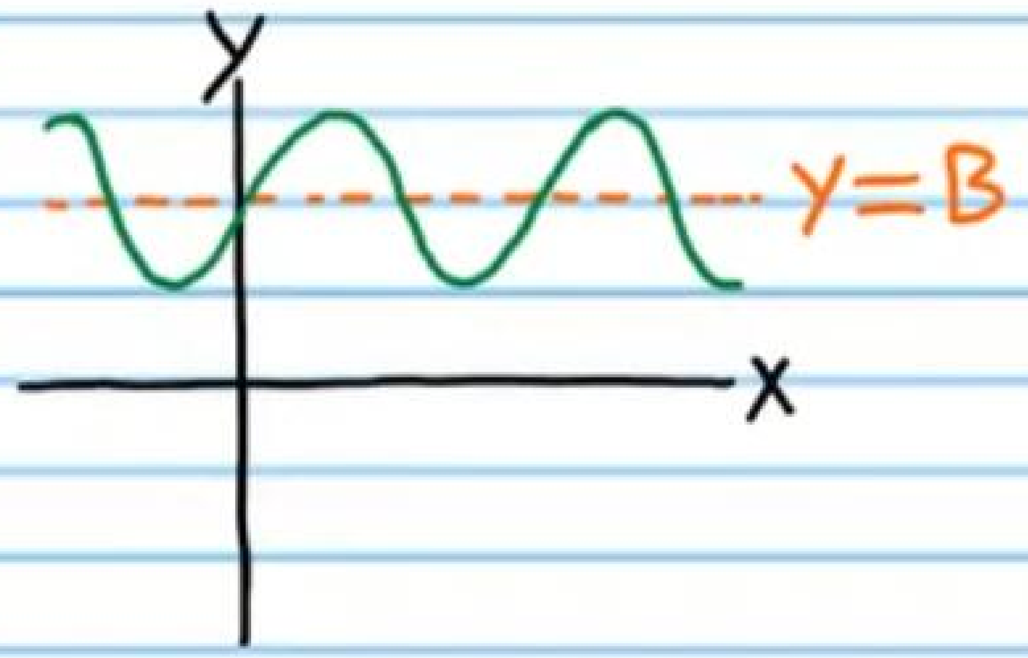
$y = A\cot(\omega x - \phi) + B$

Sinusoidal Function: A function whose graph has the same general form as the sine or cosine functions.

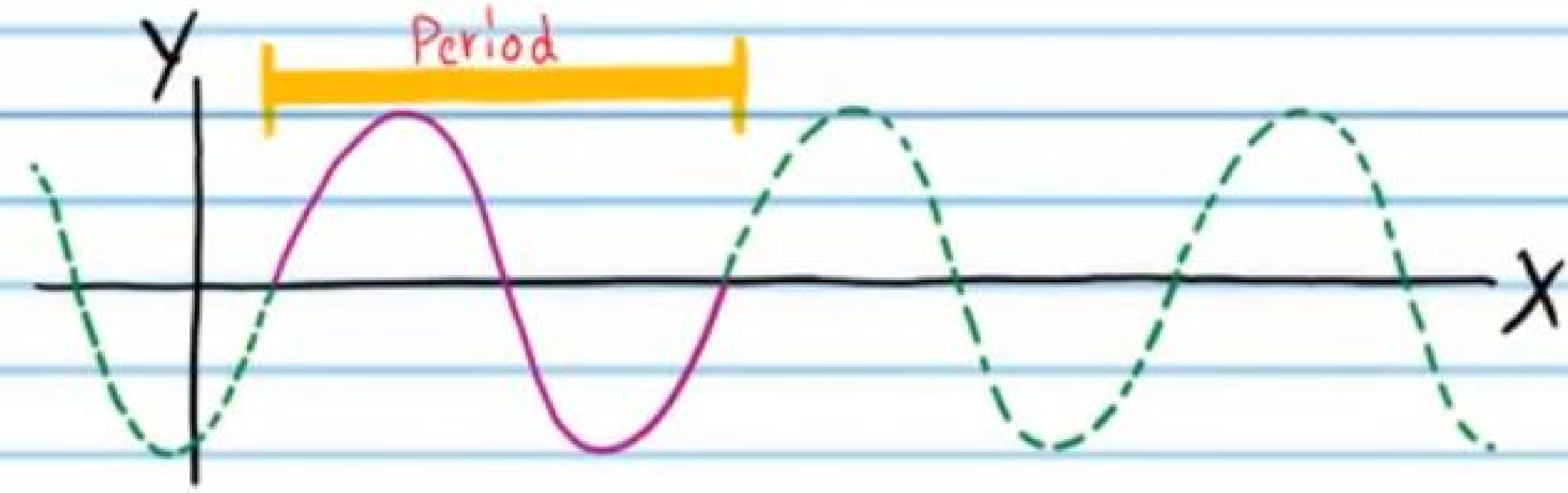
Midline: A horizontal line drawn through the middle of the graph of a trigonometric function to assist when graphing.

For the functions above, the equation of the midline is $y = B$.

Example:



Period: The number of degrees required for a trigonometric function to complete one whole cycle.



The period of the sine, cosine, cosecant, and secant functions is given by the following equation.

$$P = \frac{360^\circ}{\omega}$$

The period of the tangent and cotangent functions is given by the following equation.

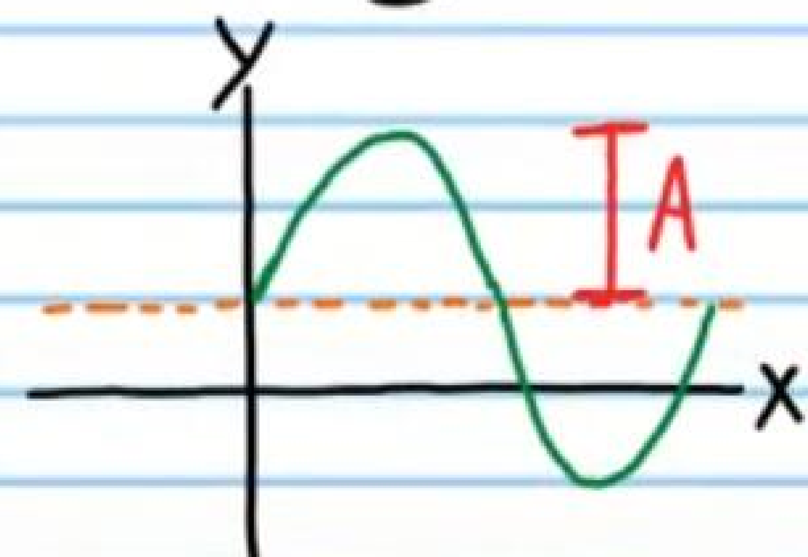
$$P = \frac{180^\circ}{\omega}$$

Phase Shift: The horizontal distance that a parent graph is translated (to the right or left).

The phase shift is given by the equation below.

$$PS = \frac{\phi}{\omega}$$

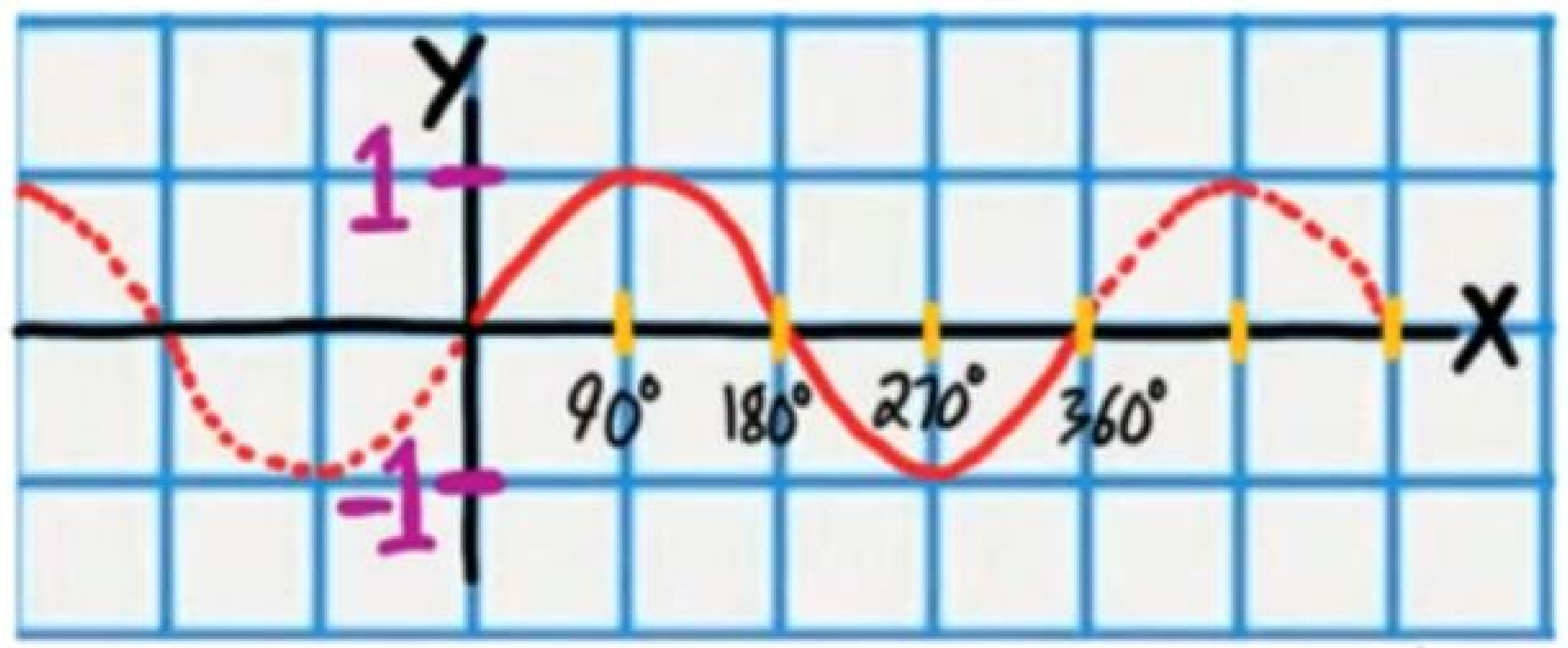
Amplitude: The vertical distance from the midline to the highest or lowest point on a sinusoidal graph.



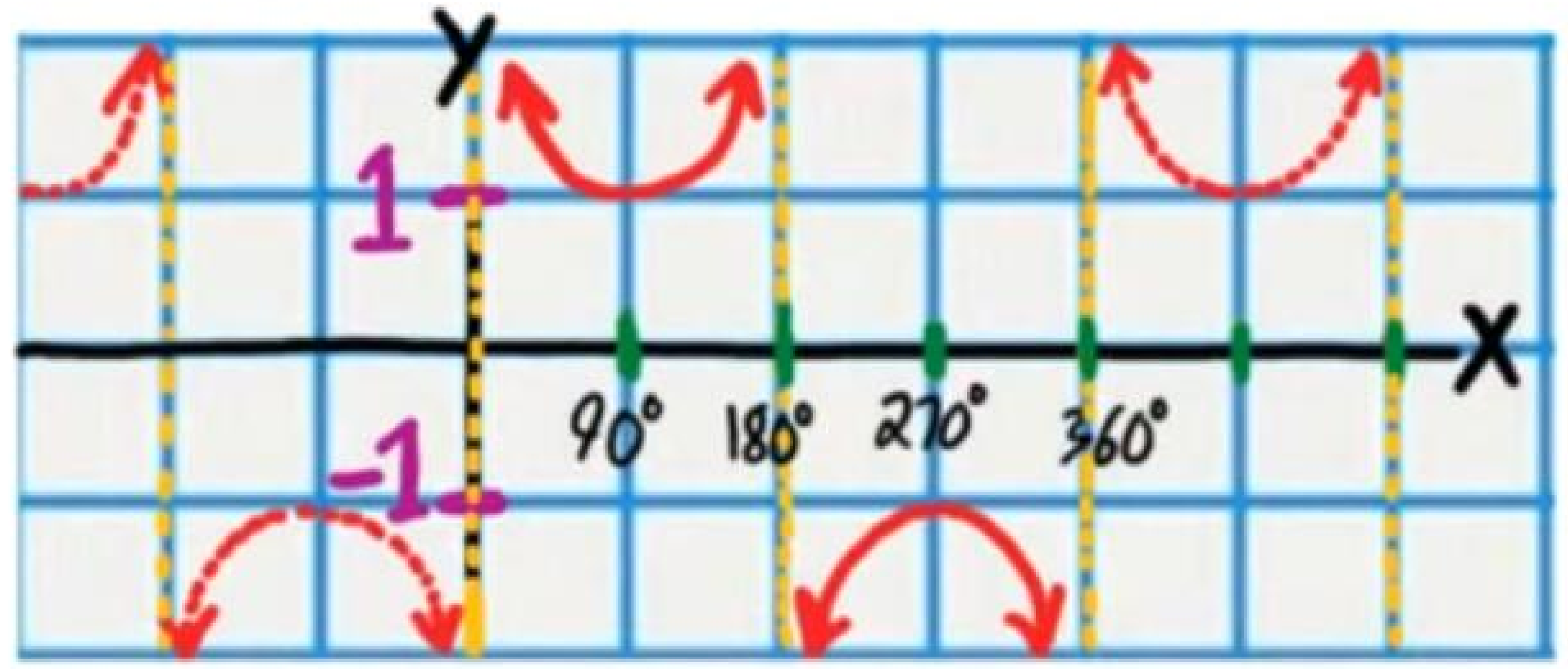
The amplitude is the magnitude (i.e. the absolute value) of the coefficient of the trigonometric function.

Example: If $y = A\sin(\omega x - \phi) + B$, then Amplitude $= |A|$.

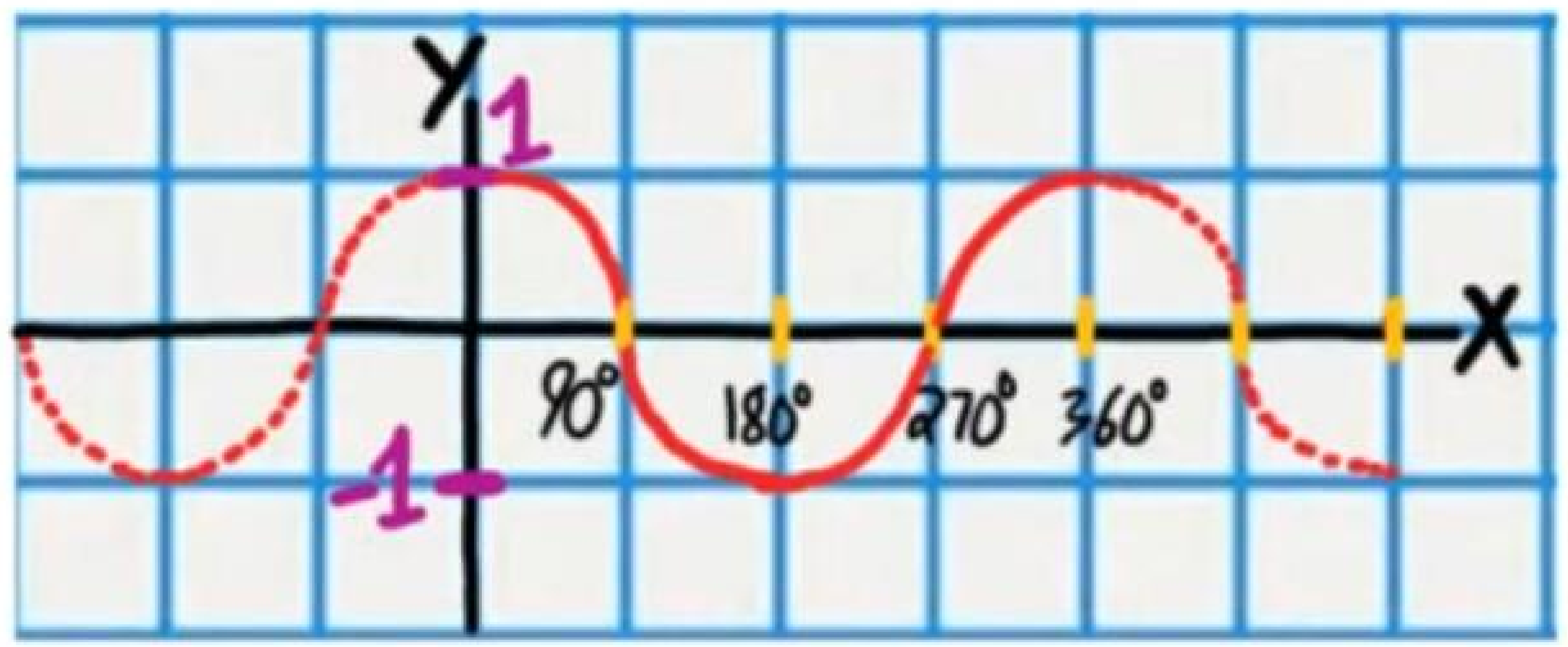
y = sinx



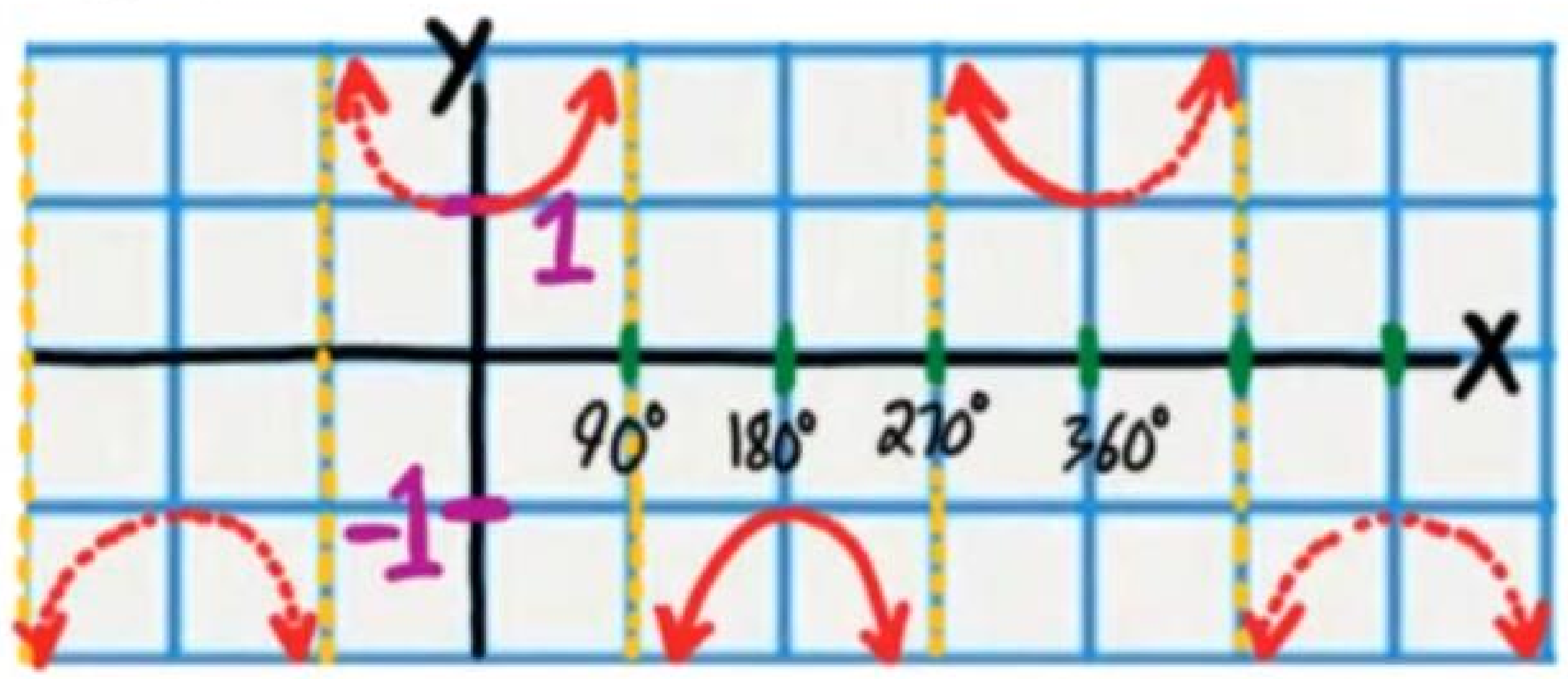
y = cscx



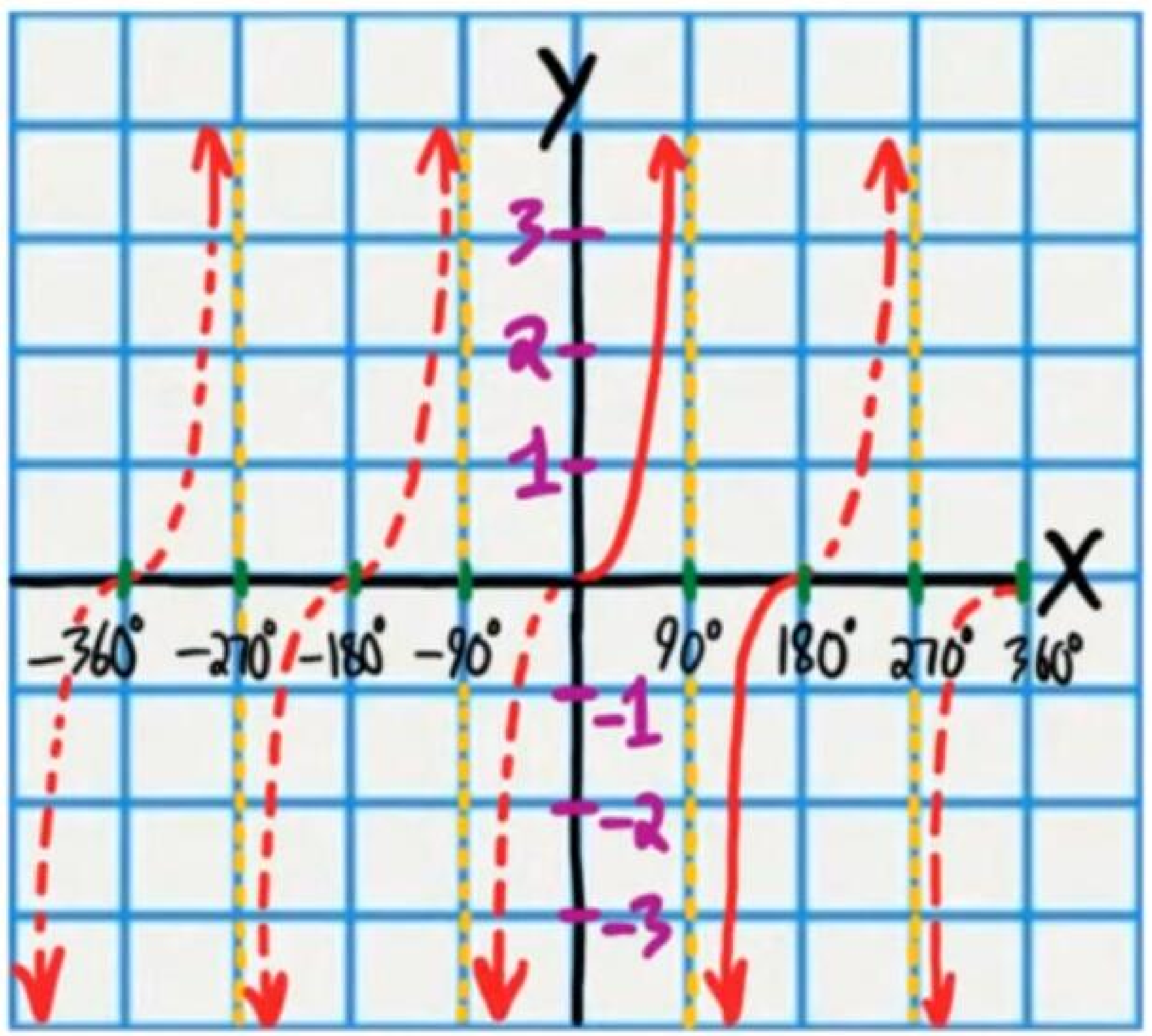
y = cosx



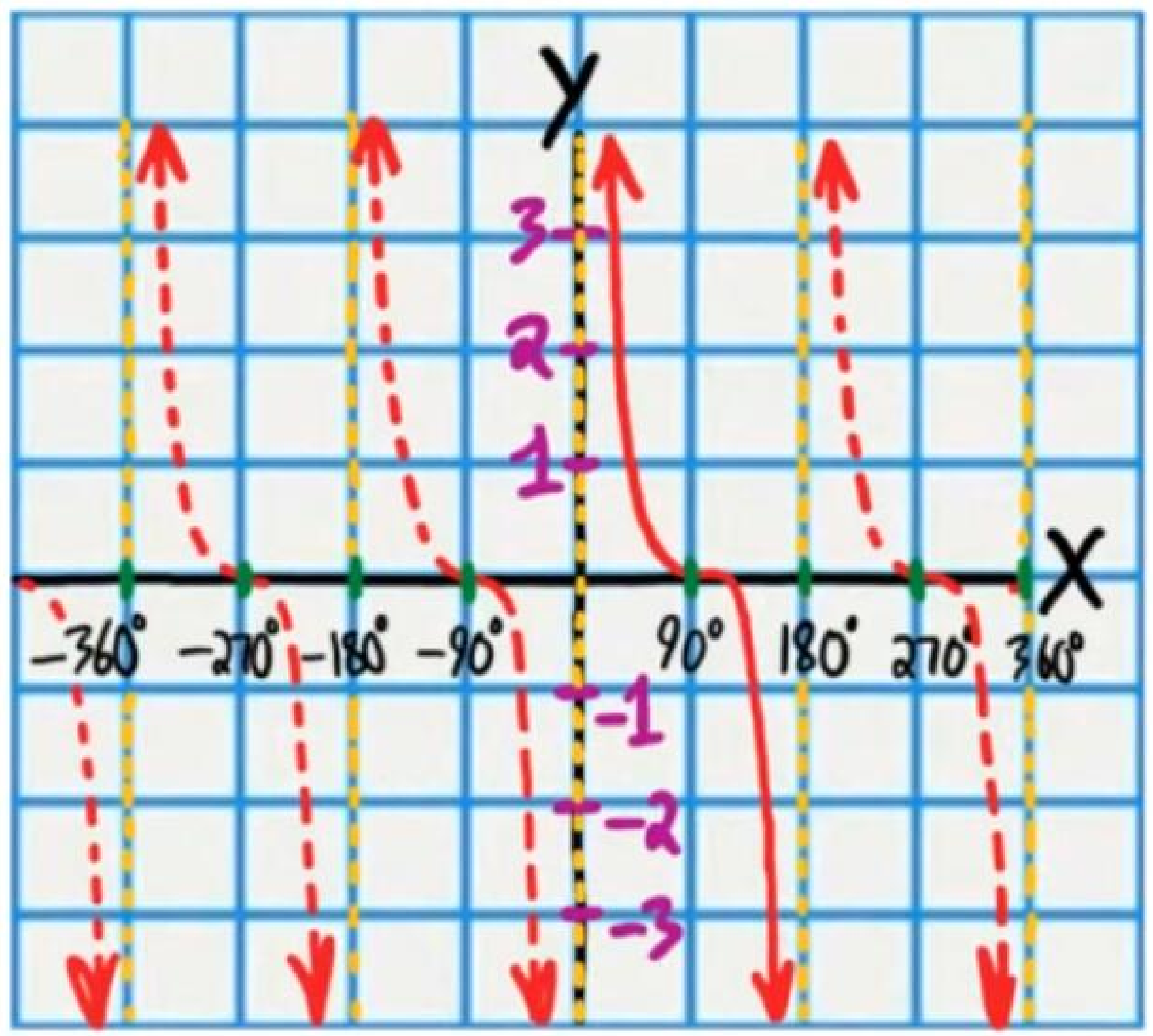
y = secx



y = tanx



y = cotx



For the following problems, write the equation of the midline, and find the period, amplitude, and phase shift. Then, graph the equation.

① $y = 2\sin(3x - 270°) - 1$

Midline Equation: $\boxed{y = -1}$

Period: $P = \frac{360°}{w} = \frac{360°}{3} = \boxed{120°}$

Phase Shift: $PS = \frac{\phi}{w} = \frac{270°}{3} = \boxed{90°}$

Amplitude: $|A| = |2| = \boxed{2}$

Change the form of the equation.

$$y = 2\sin(3x - 270°) - 1$$

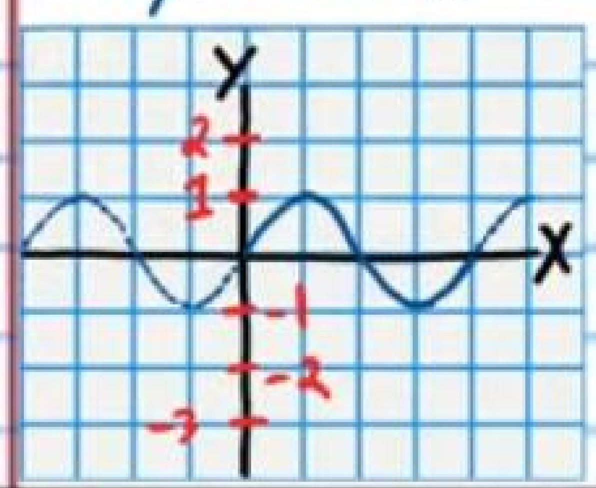
$$y + 1 = 2\sin(3x - 270°)$$

$$\frac{y+1}{2} = \sin(3x - 270°)$$

$$\frac{y+1}{2} = \sin[3(x - 90°)]$$

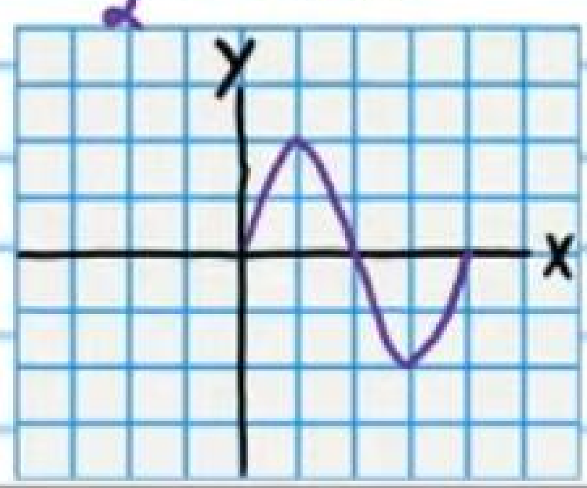
Parent Function

$y = \sin x$



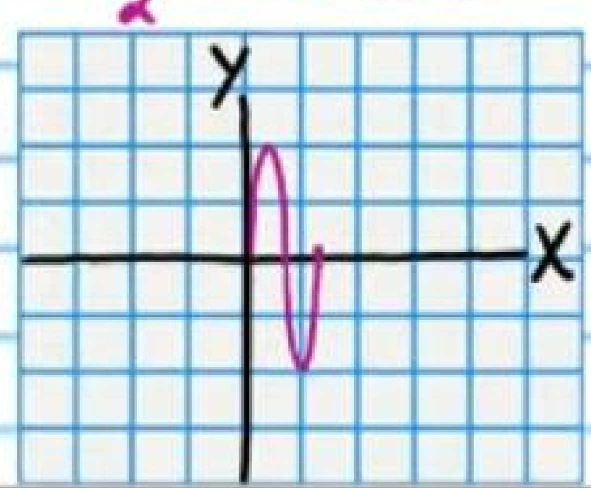
Transformation #1

$\frac{y}{2} = \sin x$



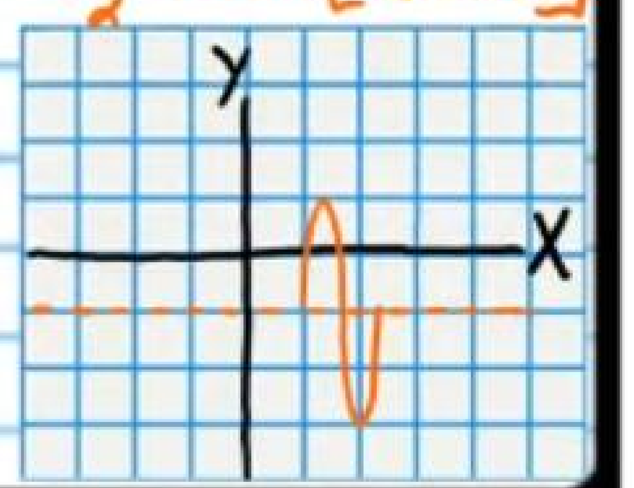
Transformation #2

$\frac{y}{2} = \sin(3x)$



Final Transformation

$\frac{y+1}{2} = \sin[3(x - 90°)]$



② $y = 3\sin(2x + 360°) + 2$

Midline Equation: $y = 2$

Period: $P = \frac{360°}{w} = \frac{360°}{2} = \boxed{180°}$

Phase Shift: $PS = \frac{\phi}{w} = \frac{-360°}{2} = -180°$

Amplitude: $|A| = |3| = 3$

Change the form of the equation.

$$y = 3\sin(2x + 360°) + 2$$

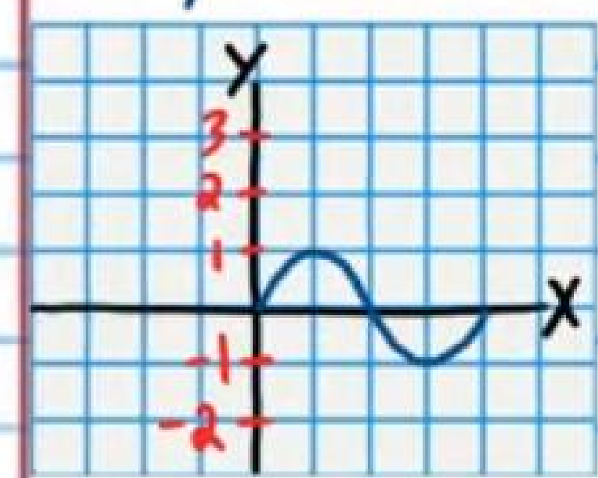
$$y - 2 = 3\sin(2x + 360°)$$

$$\frac{y-2}{3} = \sin(2x + 360°)$$

$$\frac{y-2}{3} = \sin[2(x + 180°)]$$

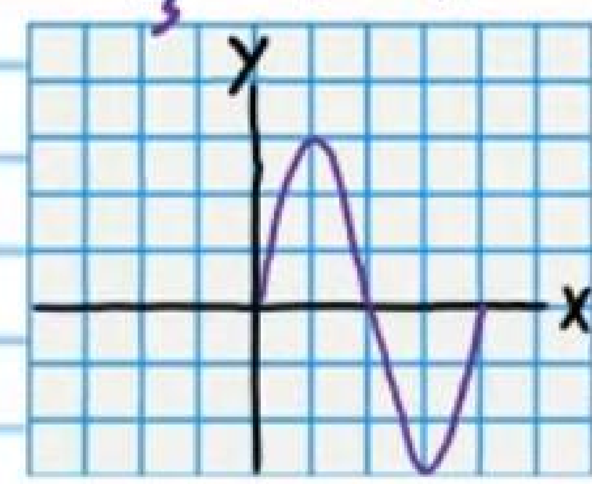
Parent Function

$y = \sin x$



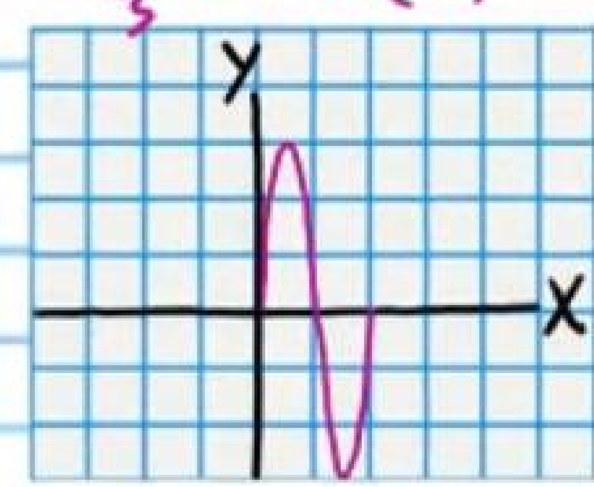
Transformation #1

$\frac{y}{3} = \sin x$



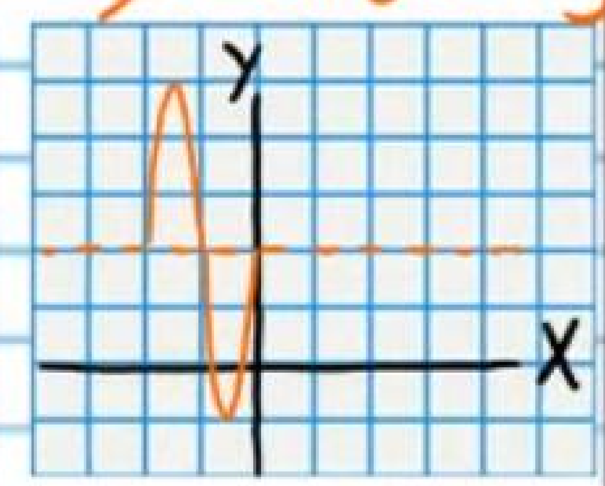
Transformation #2

$\frac{y}{3} = \sin(2x)$



Final Transformation

$\frac{y-2}{3} = \sin[2(x + 180°)]$



③ $y = 4\sin(2x - 90^\circ) + 3$

Midline Equation: $\boxed{y=3}$

Period: $P = \frac{360^\circ}{\omega} = \frac{360^\circ}{2} = \boxed{180^\circ}$

Phase Shift: $PS = \frac{\phi}{\omega} = \frac{90^\circ}{2} = \boxed{45^\circ}$

Amplitude: $|A| = |4| = \boxed{4}$

Change the form of the equation.

$$y = 4\sin(2x - 90^\circ) + 3$$

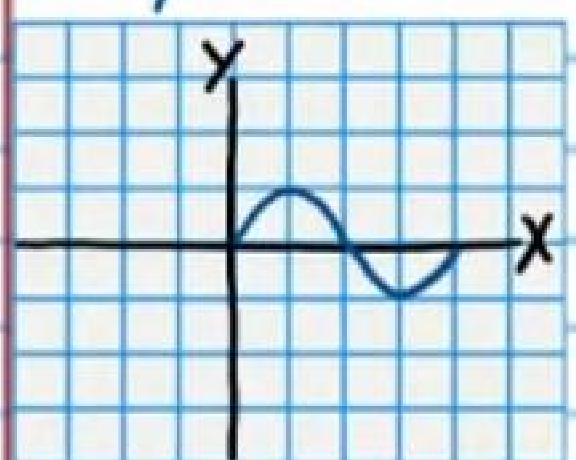
$$y - 3 = 4\sin(2x - 90^\circ)$$

$$\frac{y-3}{4} = \sin(2x - 90^\circ)$$

$$\frac{y-3}{4} = \sin[2(x - 45^\circ)]$$

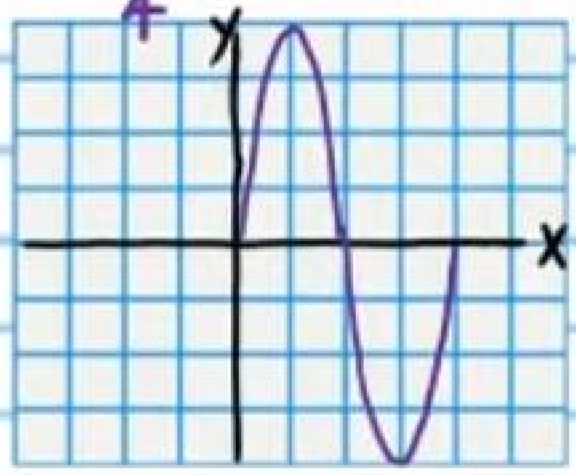
Parent Function

$y = \sin x$



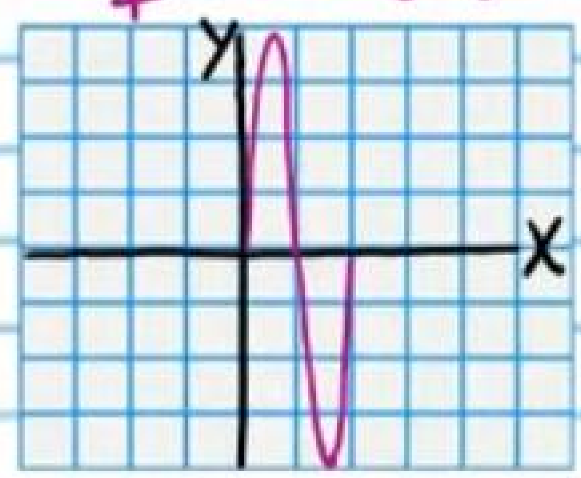
Transformation #1

$\frac{y}{4} = \sin x$



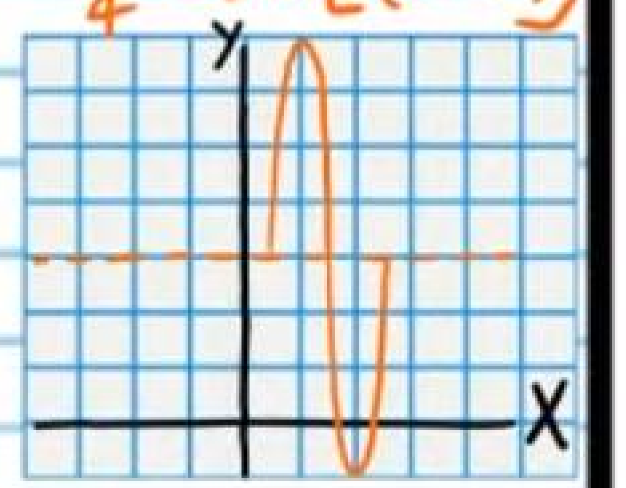
Transformation #2

$\frac{y}{4} = \sin(2x)$



Final Transformation

$\frac{y-3}{4} = \sin[2(x-45^\circ)]$



④ $y = 2\sin(3x + 360^\circ) - 2$

Midline Equation: $\boxed{y=-2}$

Period: $P = \frac{360^\circ}{\omega} = \frac{360^\circ}{3} = \boxed{120^\circ}$

Phase Shift: $PS = \frac{\phi}{\omega} = \frac{-360^\circ}{3} = \boxed{-120^\circ}$

Amplitude: $|A| = |2| = 2$

Change the form of the equation.

$$y = 2\sin(3x + 360^\circ) - 2$$

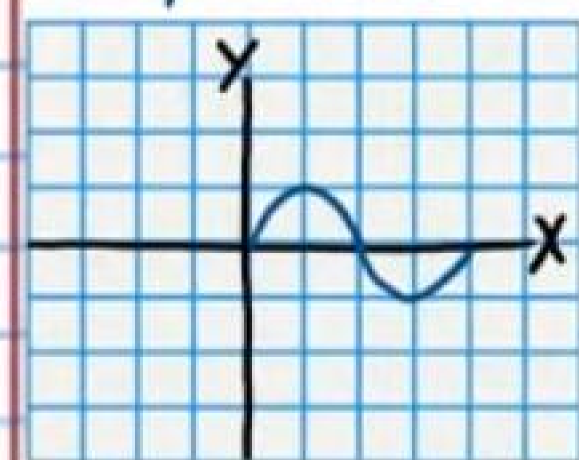
$$y + 2 = 2\sin(3x + 360^\circ)$$

$$\frac{y+2}{2} = \sin(3x + 360^\circ)$$

$$\frac{y+2}{2} = \sin[3(x + 120^\circ)]$$

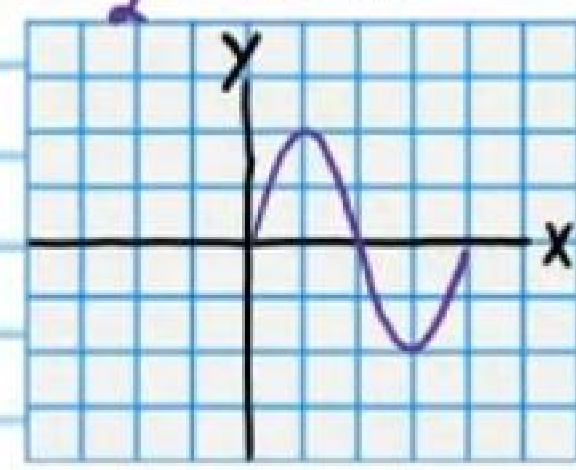
Parent Function

$y = \sin x$



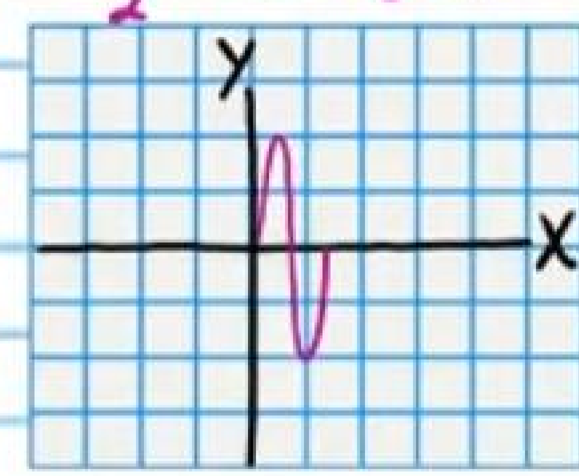
Transformation #1

$\frac{y}{2} = \sin x$



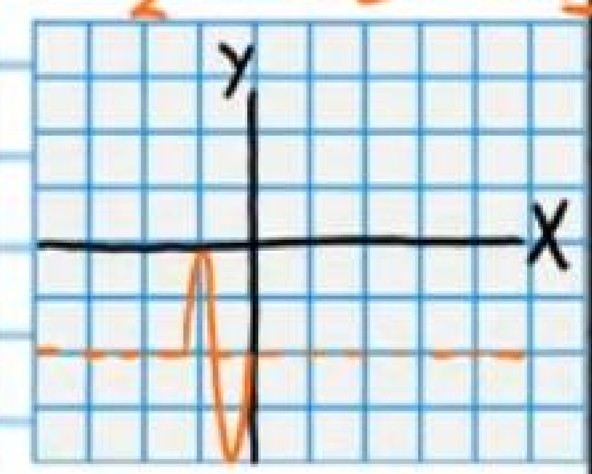
Transformation #2

$\frac{y}{2} = \sin(3x)$



Final Transformation

$\frac{y+2}{2} = \sin[3(x+120^\circ)]$



⑤ $y = -4\cos\left(\frac{1}{2}x - 180°\right) + 3$

Midline Equation: $\boxed{y = 3}$

Period: $P = \frac{360°}{\omega} = \frac{360°}{\frac{1}{2}} = 360° \div \frac{1}{2} = 360° \cdot \frac{2}{1} = \boxed{720°}$

Phase Shift: $PS = \frac{\Phi}{\omega} = \frac{180°}{\frac{1}{2}} = \boxed{360°}$

Amplitude: $|A| = |-4| = 4$

Change the form of the equation.

$y = -4\cos\left(\frac{x}{2} - 180°\right) + 3$

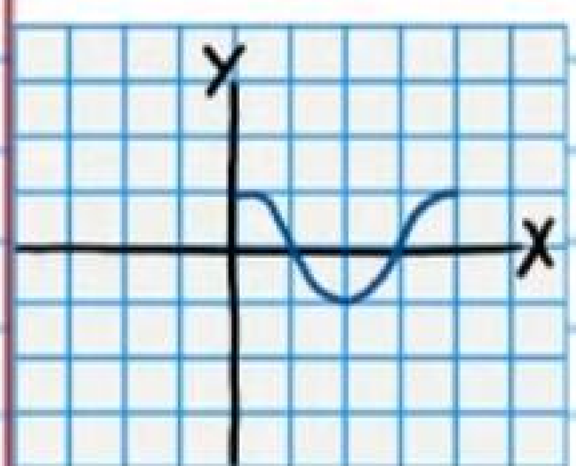
$y - 3 = -4\cos\left(\frac{x}{2} - 180°\right)$

$\frac{y-3}{-4} = \cos\left(\frac{x}{2} - 180°\right)$

$\frac{-(y-3)}{4} = \cos\left[\frac{1}{2}(x - 360°)\right]$

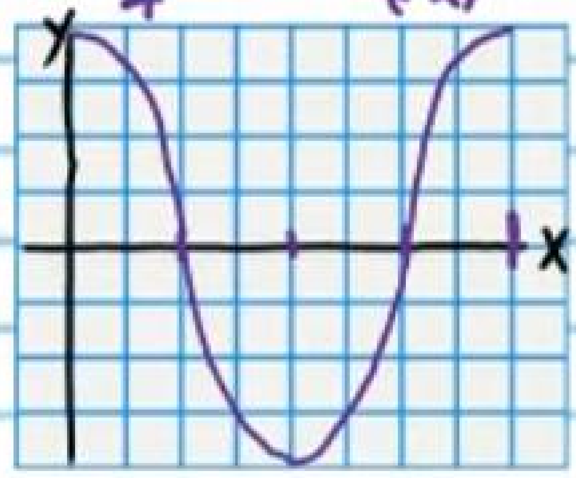
Parent Function

$y = \cos x$



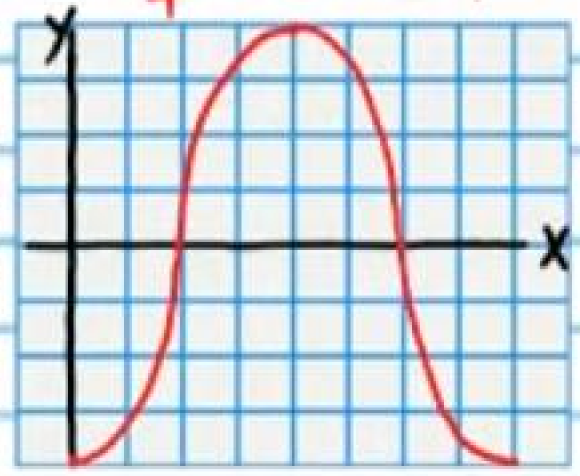
Transformation #1

$\frac{y}{4} = \cos\left(\frac{x}{2}\right)$



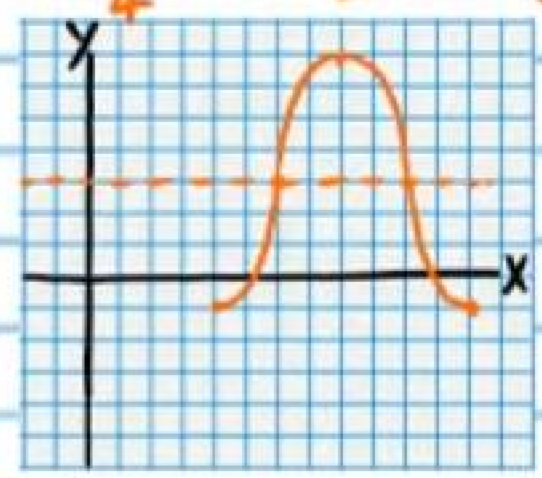
Transformation #2

$-\frac{y}{4} = \cos\left(\frac{x}{2}\right)$



Final Transformation

$\frac{-(y-3)}{4} = \cos\left[\frac{1}{2}(x - 360°)\right]$



⑥ $y = -3\cos\left(\frac{x}{3} + 45°\right) + 1$

Midline Equation: $\boxed{y = 1}$

Period: $P = \frac{360°}{\omega} = \frac{360°}{\frac{1}{3}} = 360° \div \frac{1}{3} = \boxed{1{,}080}$

Phase Shift: $PS = \frac{\Phi}{\omega} = \frac{-45°}{\frac{1}{3}} = \boxed{-135°}$

Amplitude: $|A| = |-3| = \boxed{3}$

Change the form of the equation.

$y = -3\cos\left(\frac{x}{3} + 45°\right) + 1$

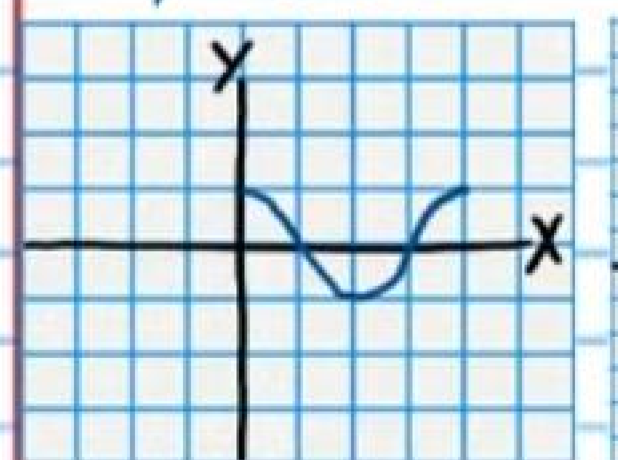
$y - 1 = -3\cos\left(\frac{x}{3} + 45°\right)$

$\frac{y-1}{-3} = \cos\left(\frac{x}{3} + 45°\right)$

$\frac{-(y-1)}{3} = \cos\left[\frac{1}{3}(x + 135°)\right]$

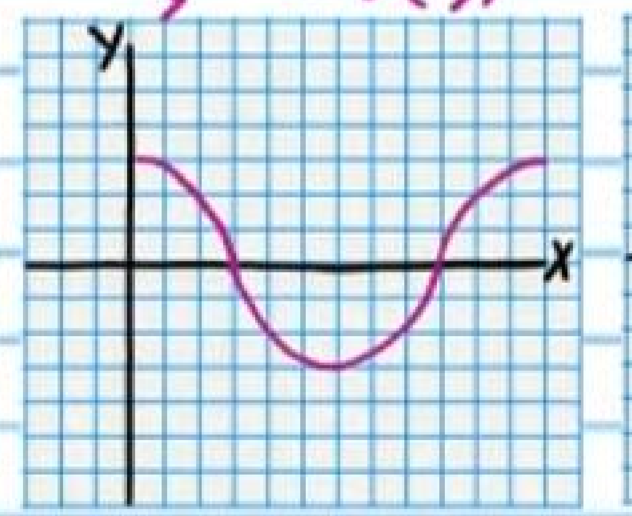
Parent Function

$y = \cos x$



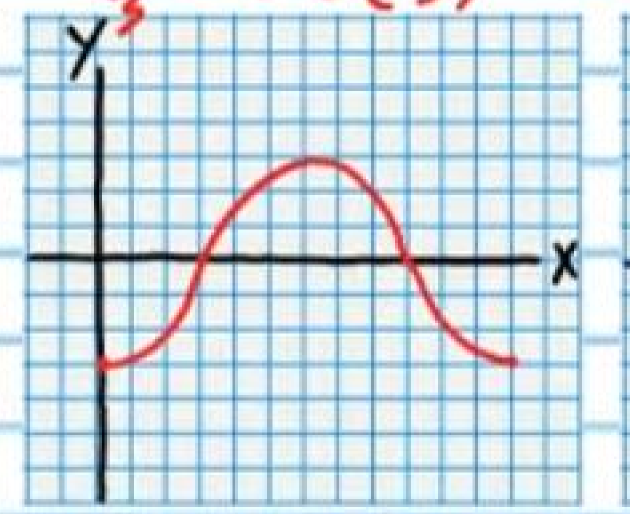
Transformation #1

$\frac{y}{3} = \cos\left(\frac{x}{3}\right)$



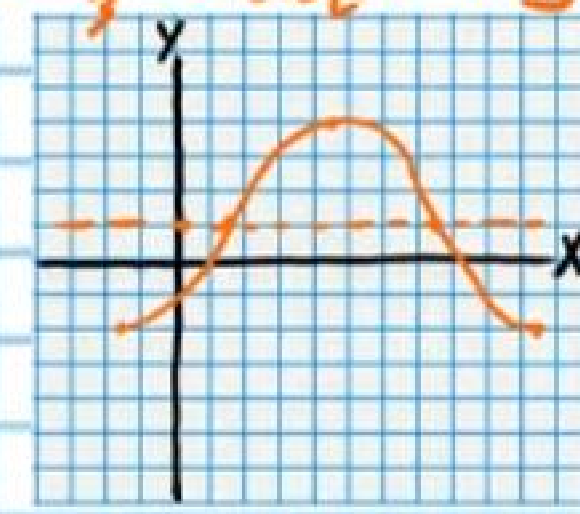
Transformation #2

$-\frac{y}{3} = \cos\left(\frac{x}{3}\right)$



Final Transformation

$\frac{-(y-1)}{3} = \cos\left[\frac{1}{3}(x + 135°)\right]$



⑦ $y = 2\csc(2x + 180°) + 1$

Midline Equation: $\boxed{y = 1}$

Period: $P = \frac{360°}{w} = \frac{360°}{2} = \boxed{180°}$

Phase Shift: $PS = \frac{\phi}{w} = \frac{-180°}{2} = \boxed{-90°}$

Change the form of the equation.

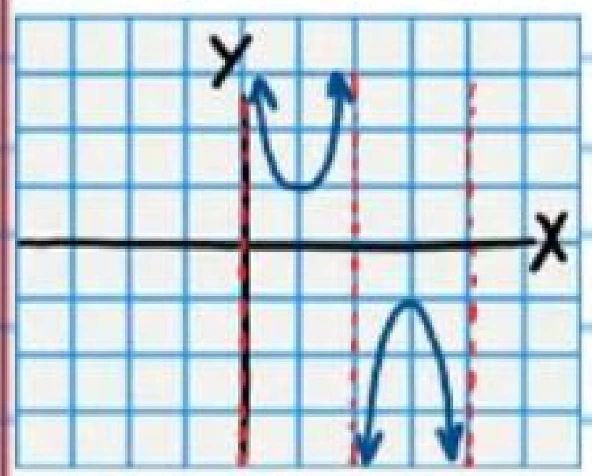
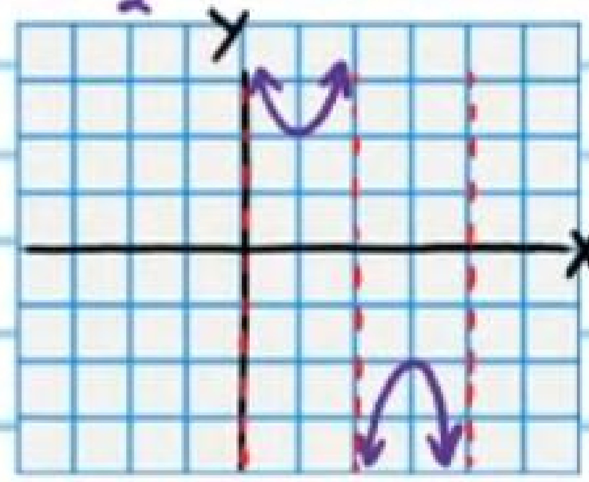
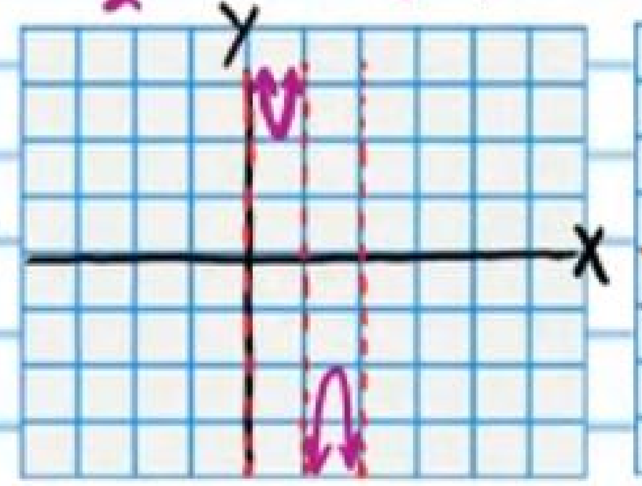
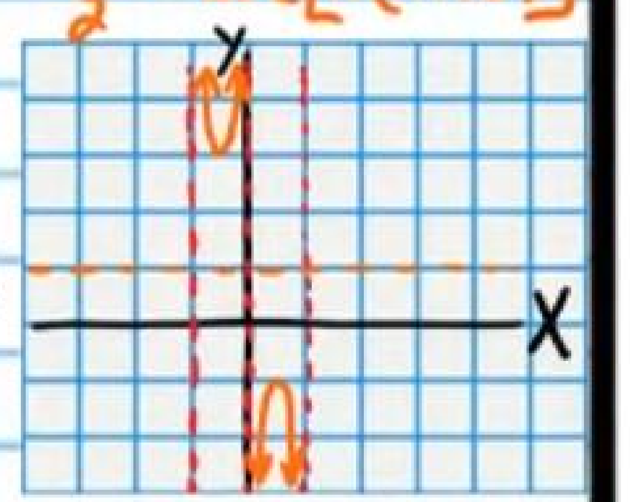
$y = 2\csc(2x + 180°) + 1$

$y - 1 = 2\csc(2x + 180°)$

$\frac{y-1}{2} = \csc(2x + 180°)$

$\frac{y-1}{2} = \csc[2(x + 90°)]$

Parent Function	Transformation #1	Transformation #2	Final Transformation
$y = \csc x$	$\frac{y}{2} = \csc x$	$\frac{y}{2} = \csc(2x)$	$\frac{y-1}{2} = \csc[2(x+90°)]$

⑧ $y = 2\sec(3x + 270°) - 1$

Midline Equation: $\boxed{y = -1}$

Period: $P = \frac{360°}{w} = \frac{360°}{3} = \boxed{120°}$

Phase Shift: $PS = \frac{\phi}{w} = \frac{-270°}{3} = \boxed{-90°}$

Change the form of the equation.

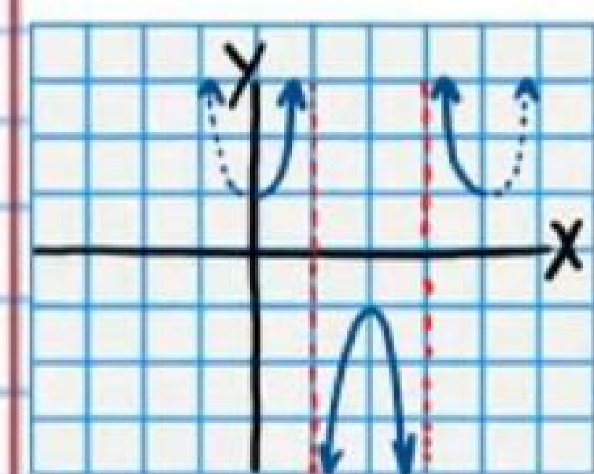
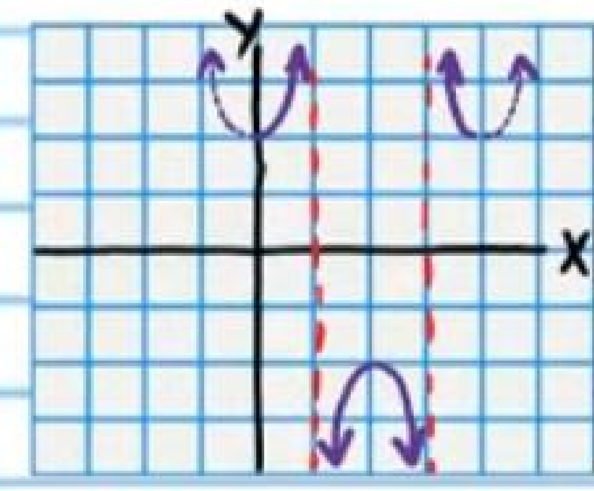
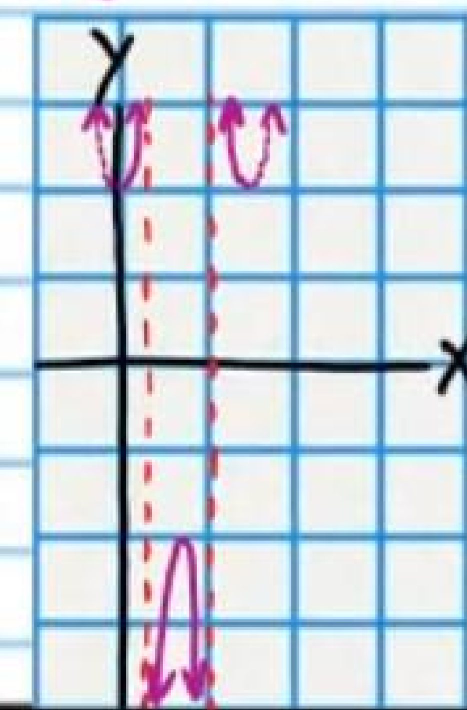
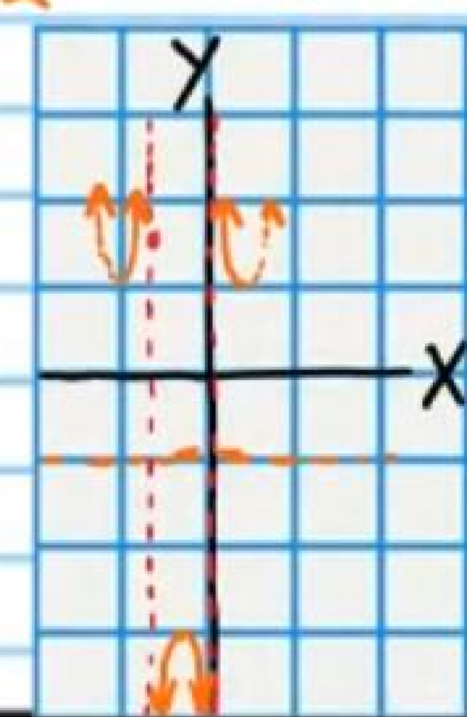
$y = 2\sec(3x + 270°) - 1$

$y + 1 = 2\sec(3x + 270°)$

$\frac{y+1}{2} = \sec(3x + 270°)$

$\frac{y+1}{2} = \sec[3(x + 90°)]$

Parent Function	Transformation #1	Transformation #2	Final Transformation
$y = \sec x$	$\frac{y}{2} = \sec x$	$\frac{y}{2} = \sec(3x)$	$\frac{y+1}{2} = \sec[3(x+90°)]$

⑨ $y = \cot\left(\frac{x}{3} - 90°\right) - 2$

Midline Equation: $\boxed{y = -2}$

Period: $P = \frac{180°}{w} = \frac{180°}{\frac{1}{3}} = \boxed{540°}$

Phase Shift: $PS = \frac{\Phi}{w} = \frac{90°}{\frac{1}{3}} = \boxed{270°}$

Change the form of the equation.

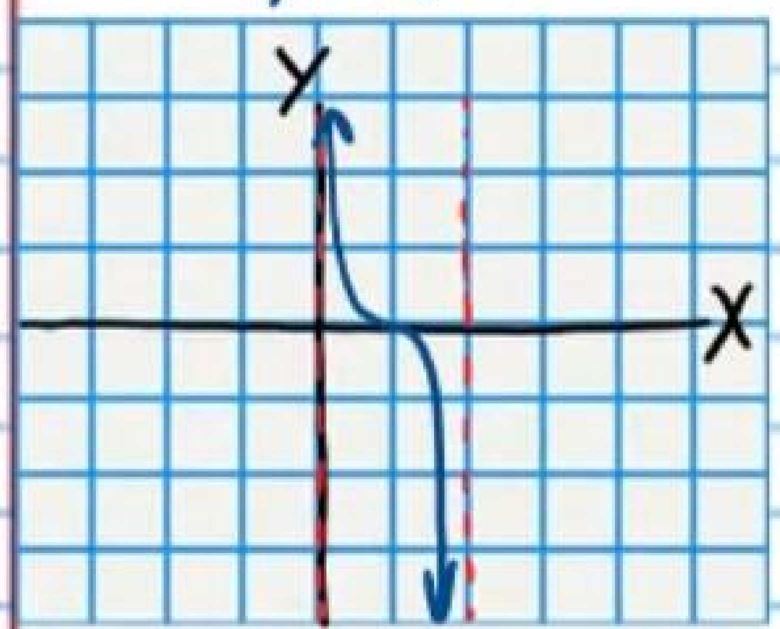
$y = \cot\left(\frac{x}{3} - 90°\right) - 2$

$y + 2 = \cot\left(\frac{x}{3} - 90°\right)$

$y + 2 = \cot\left[\frac{1}{3}(x - 270°)\right]$

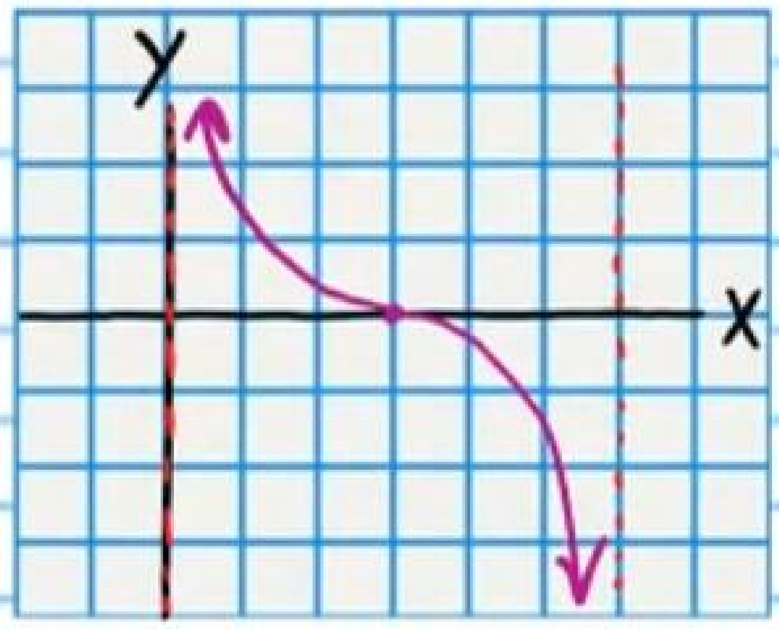
Parent Function

$y = \cot x$



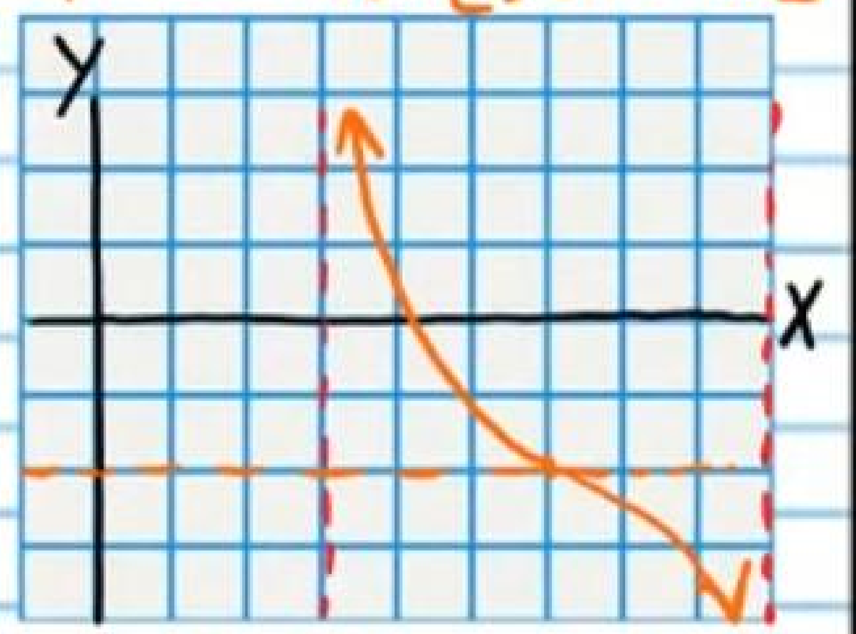
Transformation #1

$y = \cot\left(\frac{x}{3}\right)$



Final Transformation

$y + 2 = \cot\left[\frac{1}{3}(x - 270°)\right]$



⑩ $y = \tan\left(\frac{x}{2} - 90°\right) + 2$

Midline Equation: $\boxed{y = 2}$

Period: $P = \frac{180°}{w} = \frac{180°}{\frac{1}{2}} = \boxed{360°}$

Phase Shift: $PS = \frac{\Phi}{w} = \frac{90°}{\frac{1}{2}} = \boxed{180°}$

Change the form of the equation.

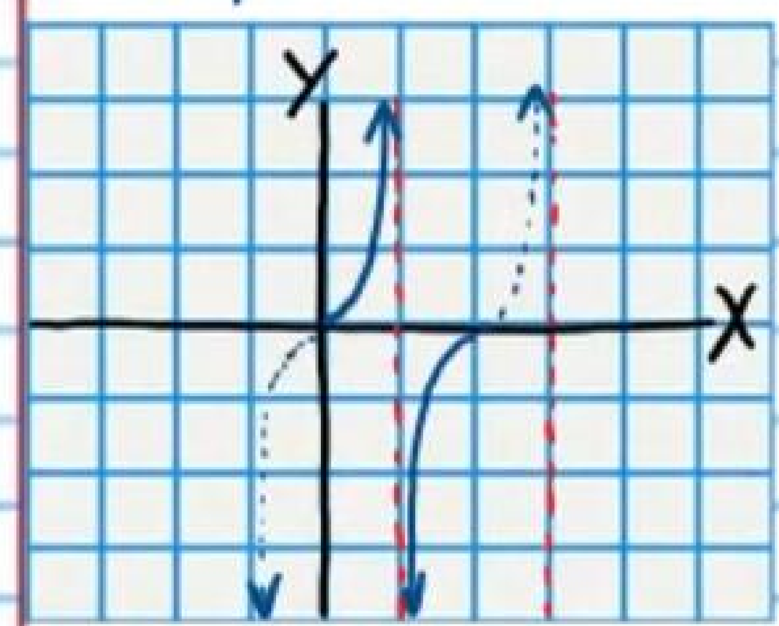
$y = \tan\left(\frac{x}{2} - 90°\right) + 2$

$y - 2 = \tan\left(\frac{x}{2} - 90°\right)$

$y - 2 = \tan\left[\frac{1}{2}(x - 180°)\right]$

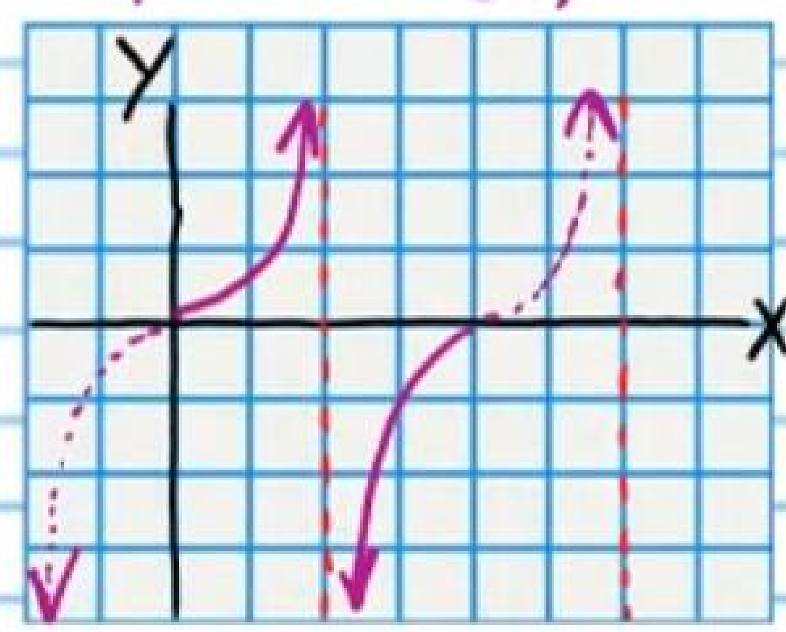
Parent Function

$y = \tan x$



Transformation #1

$y = \tan\left(\frac{x}{2}\right)$



Final Transformation

$y - 2 = \tan\left[\frac{1}{2}(x - 180°)\right]$

Name
Date
Course

Transforming the Graphs of Trigonometric Functions (Homework)

For the following problems, write the equation of the midline, and find the period, amplitude, and phase shift. Then, graph the equation.

① $y = 3\cos(2x + 360°) - 2$

Midline Equation:

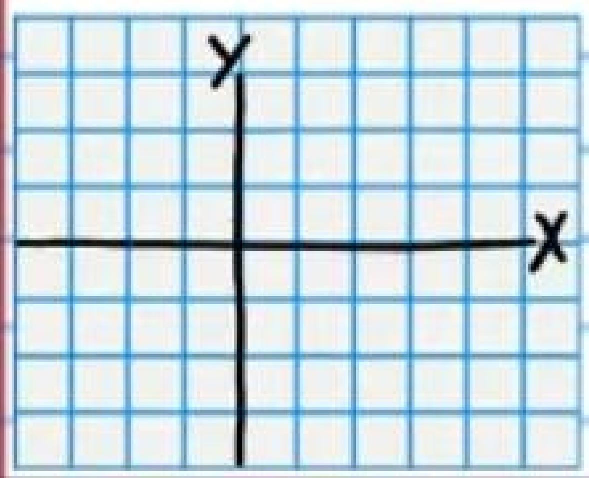
Period:

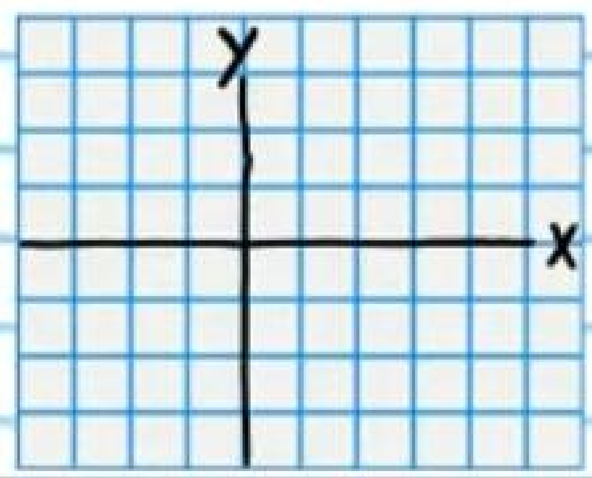
Phase Shift:

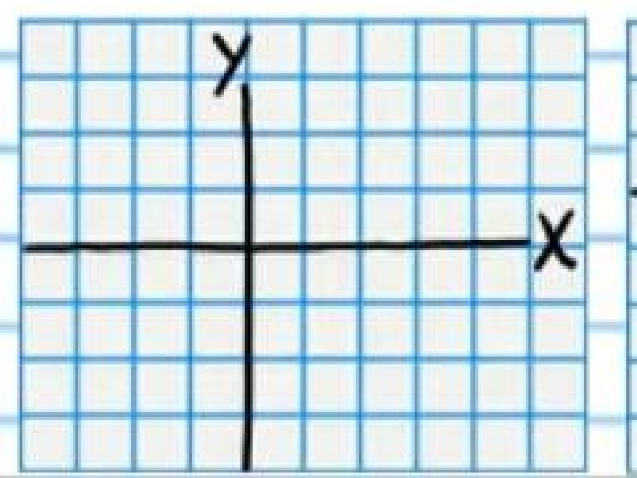
Amplitude:

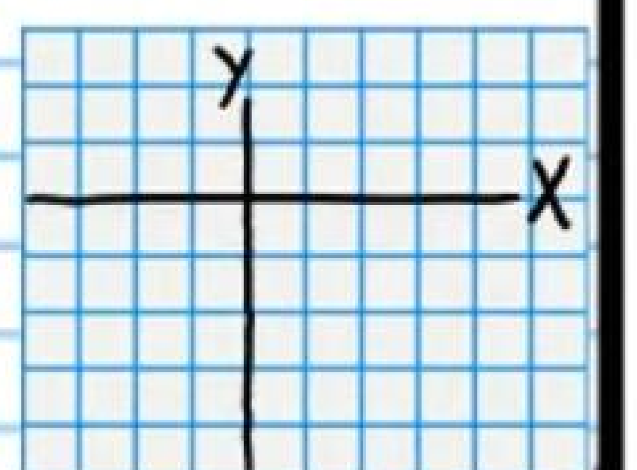
Change the form of the equation.

Parent Function	Transformation #1	Transformation #2	Final Transformation









② $y = -2\sin(3x - 270°) + 3$

Midline Equation:

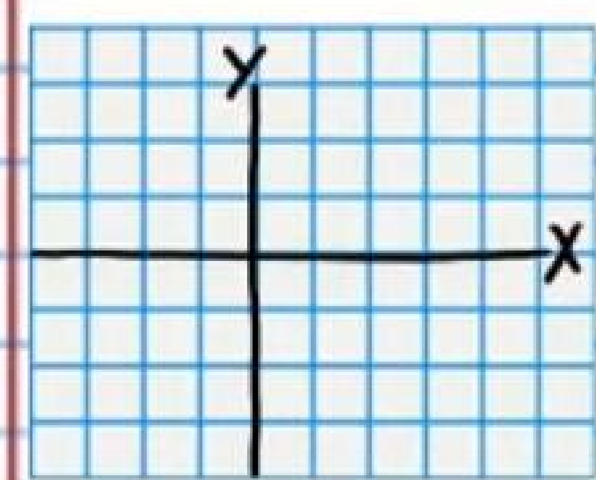
Period:

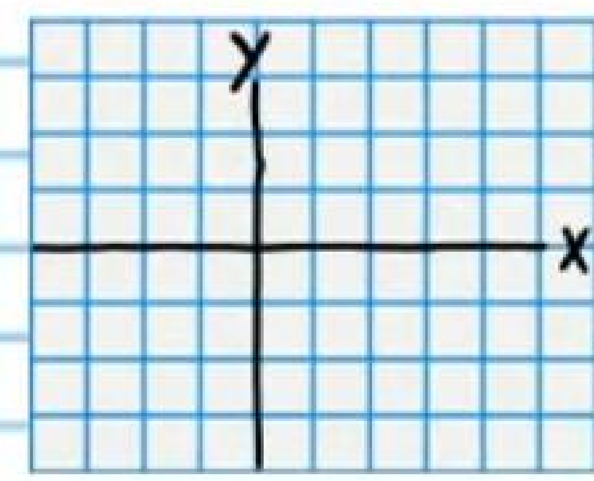
Phase Shift:

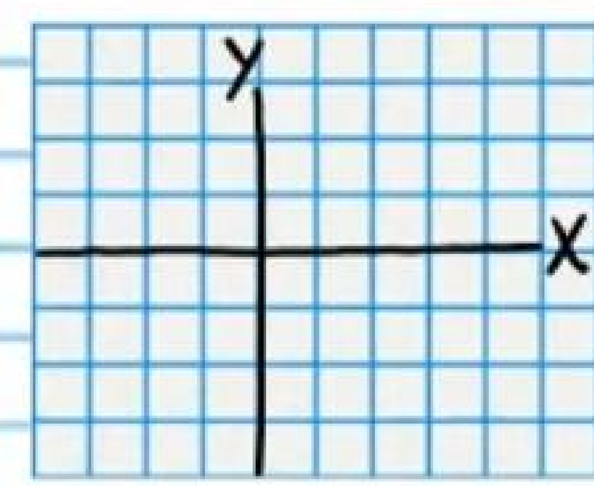
Amplitude:

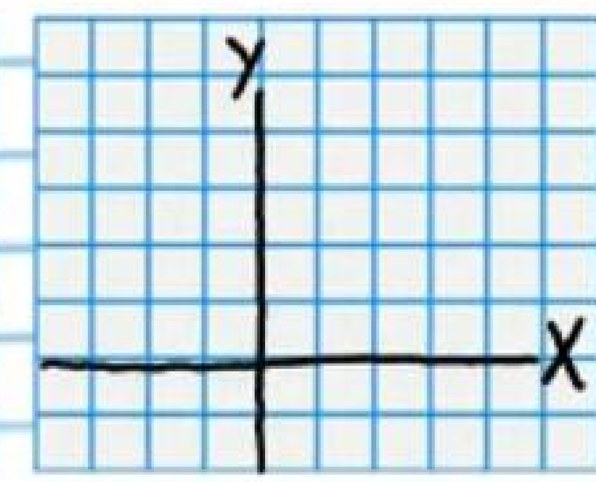
Change the form of the equation.

Parent Function	Transformation #1	Transformation #2	Final Transformation









③ $y = -\cos\left(\frac{x}{2} - 45°\right) - 1$

Midline Equation:

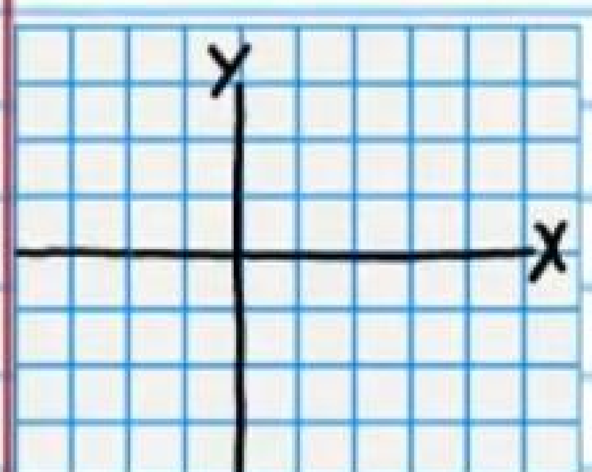
Period:

Phase Shift:

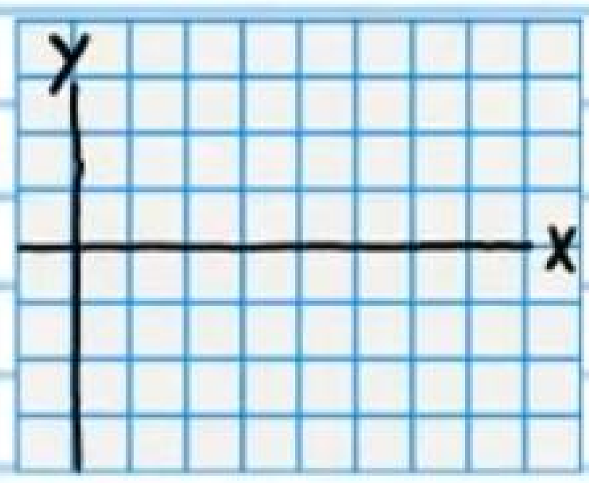
Amplitude:

Change the form of the equation.

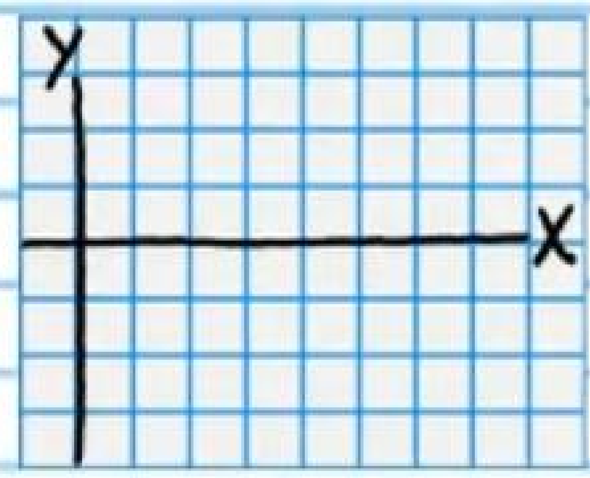
Parent Function



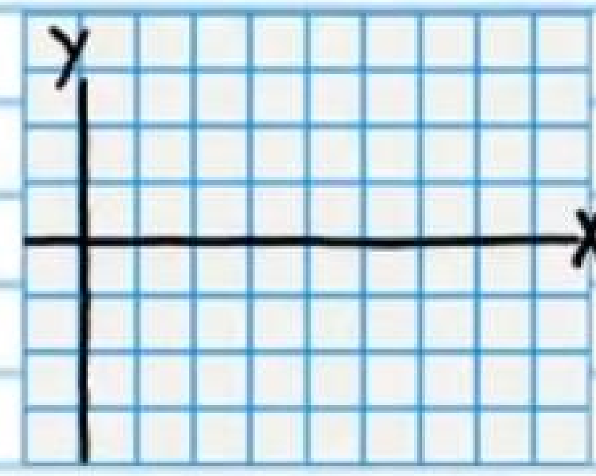
Transformation #1



Transformation #2



Final Transformation



④ $y = 4\csc\left(\frac{x}{3} + 45°\right) + 1$

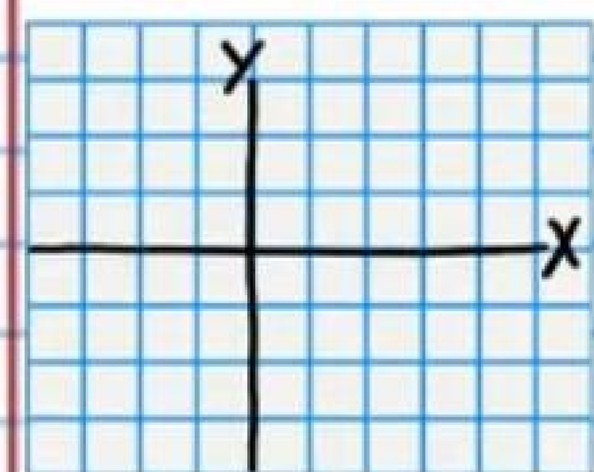
Midline Equation:

Period:

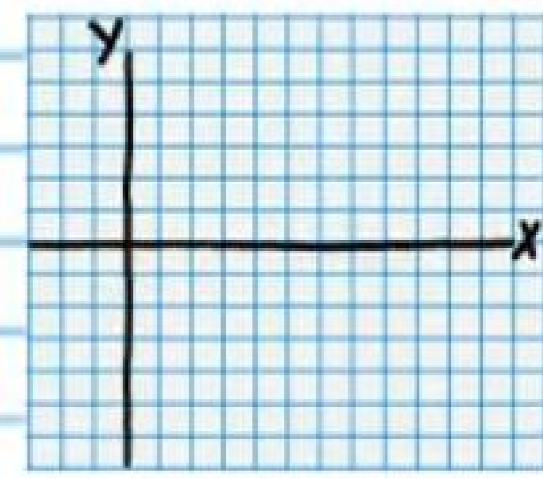
Phase Shift:

Change the form of the equation.

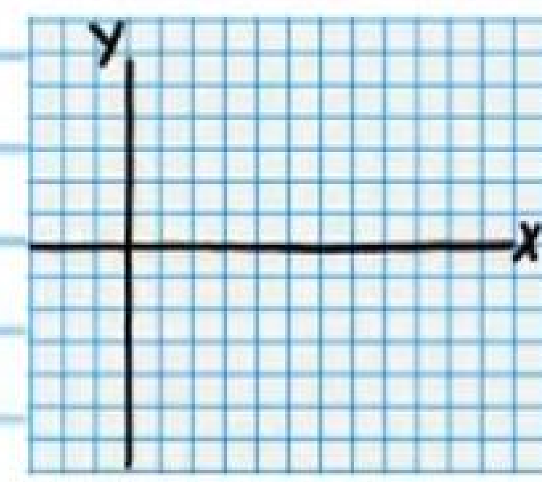
Parent Function



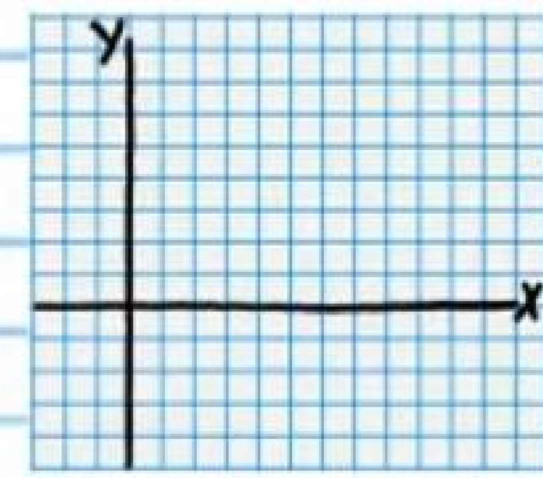
Transformation #1



Transformation #2



Final Transformation



⑤ $y = -3\sec\left(\frac{x}{2} - 135^\circ\right)$

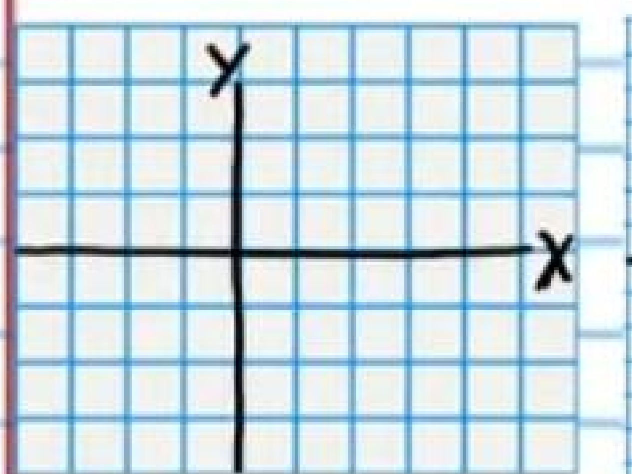
Midline Equation:

Period:

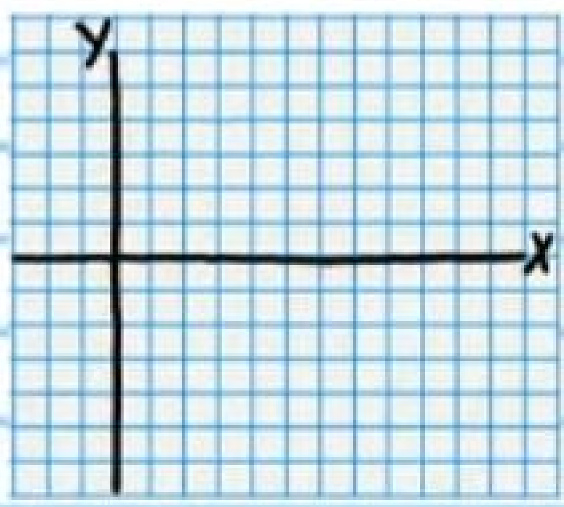
Phase Shift:

Change the form of the equation.

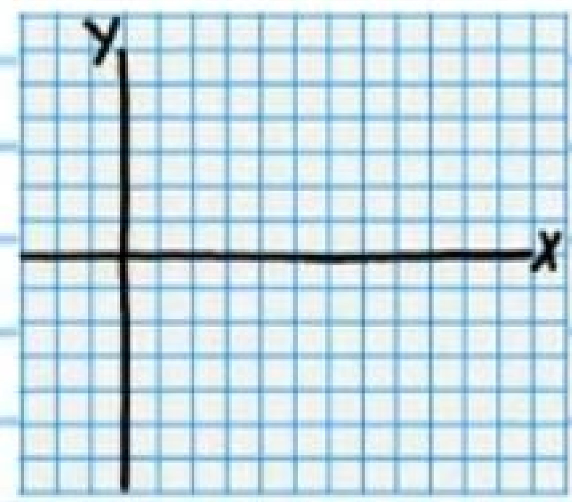
Parent Function



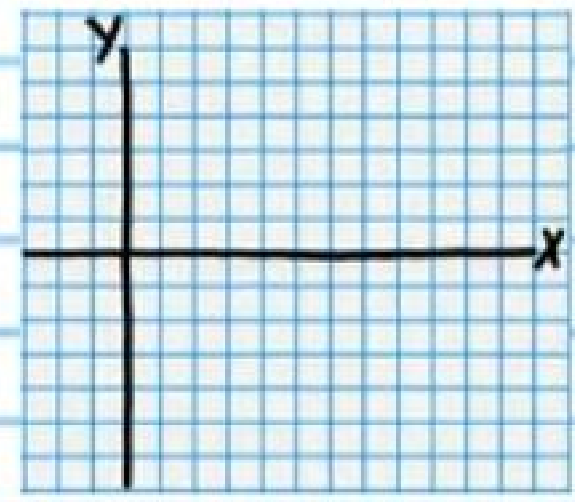
Transformation #1



Transformation #2



Final Transformation



⑥ $y = 2\tan(x + 180^\circ) - 3$

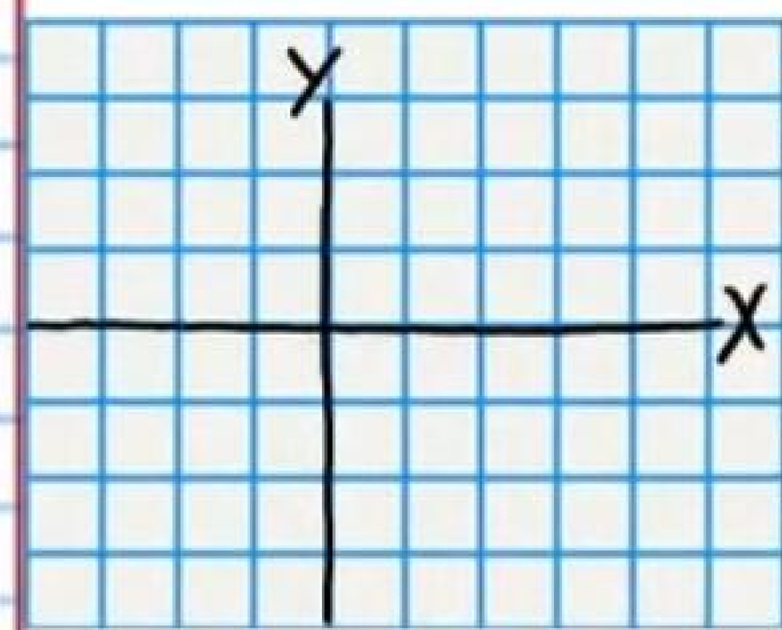
Midline Equation:

Period:

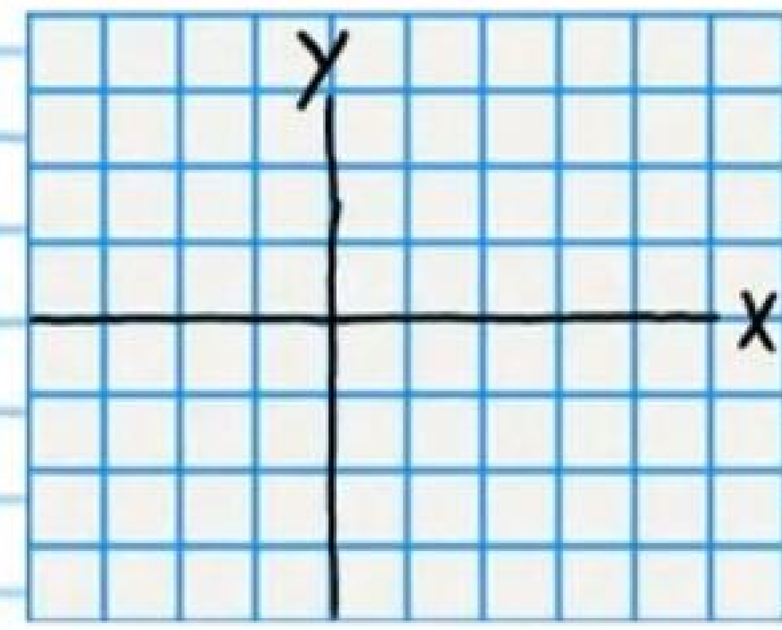
Phase Shift:

Change the form of the equation.

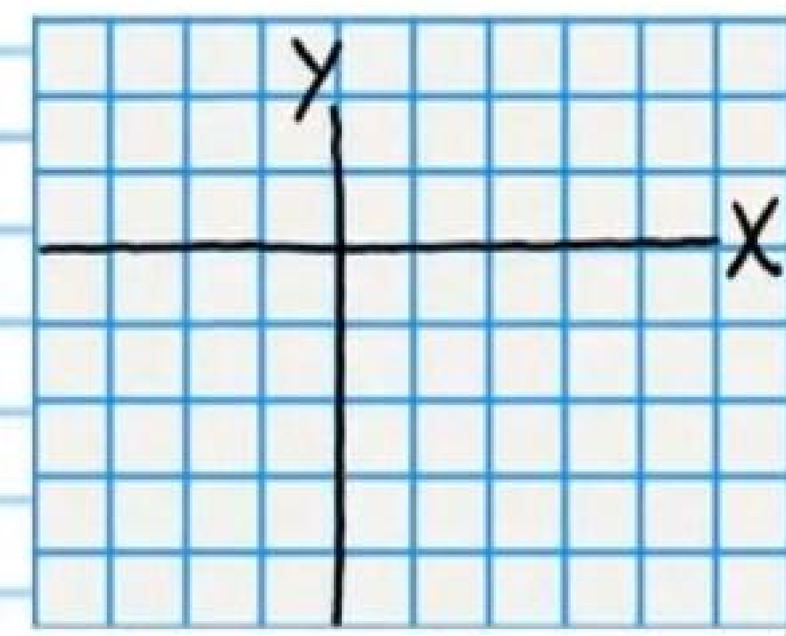
Parent Function



Transformation #1



Final Transformation



⑦ $y = \cot\left(\frac{x}{3} - 90^\circ\right)$

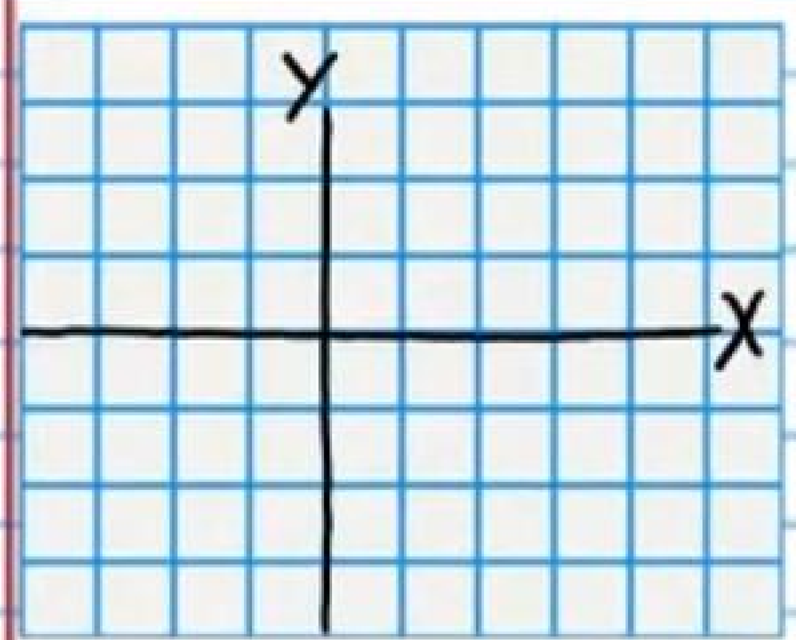
Midline Equation:

Period:

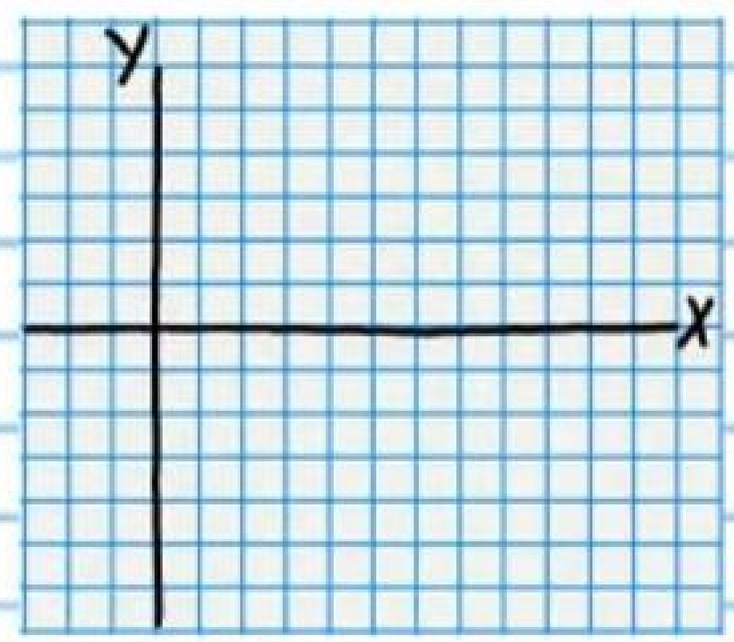
Phase Shift:

Change the form of the equation.

Parent Function



Transformation #1



Final Transformation